

EFY Note

The source code of this project is available for free download at source.efymag.com

The major functions used in the program for Arduino (*spdm_matlab.ino*) are:

- *Serial.begin(9600)* that initialises the serial port with a baud rate of 9600.
- *attachInterrupt(0,counter,FALLING)* that calls the interrupt handler 'counter' whenever a signal connected to interrupt pin (INT0) of the Arduino makes a high-to-low, that is, falling-edge transition.

The GUI application program

(*speedmeasure.m*) is developed using R2014a version of MATLAB. The 'Legacy MATLAB and Simulink Support for Arduino package' must be installed. The procedure for installation of the package is described in detail in 'Simulink Model For Real-Time Logging Of Process Data In MATLAB Workspace' DIY project published in September 2020 issue, which is also available on EFY website at <https://www.electronicsforu.com/electronics-projects/simulink-model-real-time-logging-process-data>. The PC must have MS Excel or equivalent application installed prior to running this program.

Follow the steps below:

1. Edit the COM port number in the line number 85: `a = serial('com3','BaudRate',9600);` with the corre-

sponding port number in your PC where the Arduino board is connected. You can also change the MS Excel file name and location from line number 108: `xlswrite('D:\spdm2.xlsx','z','Sheet1');`

2. Open the MATLAB program and click on the 'Run' button.

3. On pressing 'START' button in the GUI, MATLAB will try to establish communication with the Arduino. A screenshot of the GUI is shown in Fig. 4.

4. After a successful connection is established by MATLAB, you can click the 'Connect, Acquire and Display' button. MATLAB will start acquiring the speed sensor data from Arduino.

At the end of data acquisition, the program stores/saves these sample values column-wise in an Excel sheet and also displays these in the graphical window of MATLAB. A screenshot of the speed (in rpm) data logged in MS Excel is shown in Fig. 5. The plot of motor speed with respect to each sample acquired by the Arduino is shown in Fig. 6. **EFY**



Fig. 4: Screenshot of the GUI

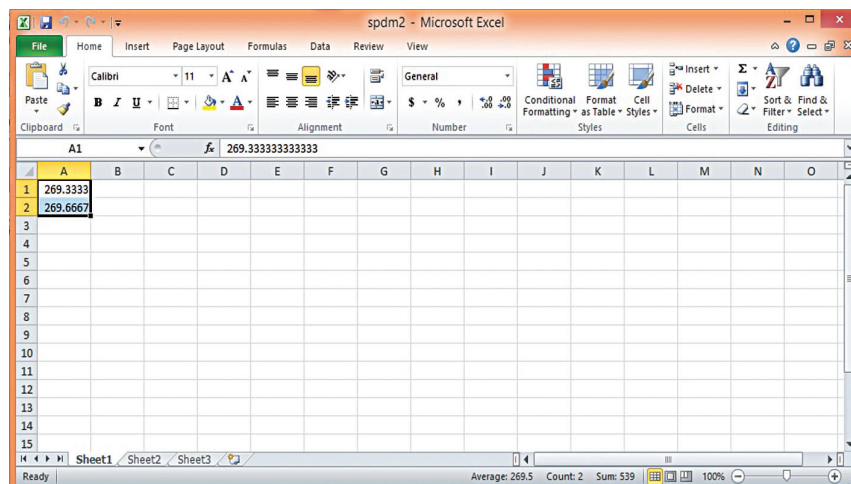


Fig. 5: Screenshot of MS Excel showing sensor data

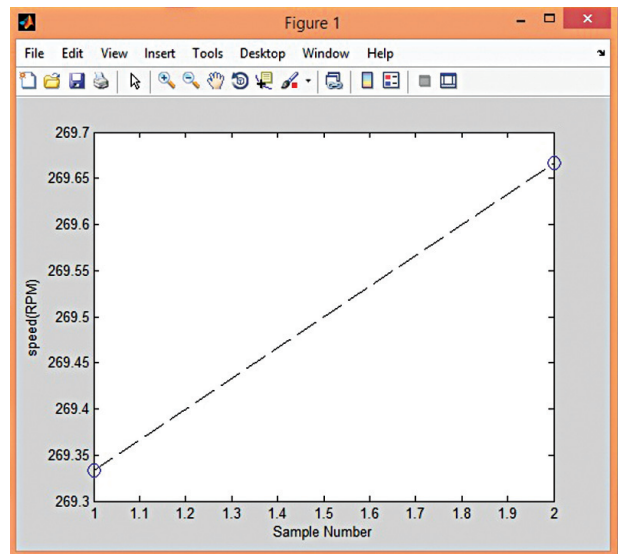


Fig. 6: MATLAB plot of motor speed (RPM) vs sample number

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